TinkerSub KB-1 side plates: cutting templates for one 24 x 24 in sheet of 1/2 in HDPE

1 Print every page at 100 % / Actual size, NOT 'Scale to fit'. Measure the 100 mm bar on a page: it must be exactly 100 mm (about 3 15/16 in).

2 Each plate is several tiles (rows A, B, C from the top, columns 1, 2, 3 from the left). Trim a tile on its grey dashed line, overlap the next tile, line up the red circles and tape the joints on the front.

3 Lay both templates on the sheet as shown on the right, 5 mm apart (the sheet has only ~2 mm to spare at each edge: measure it first). Spray adhesive (3M Super 77) or glue stick holds them; peel off afterwards.

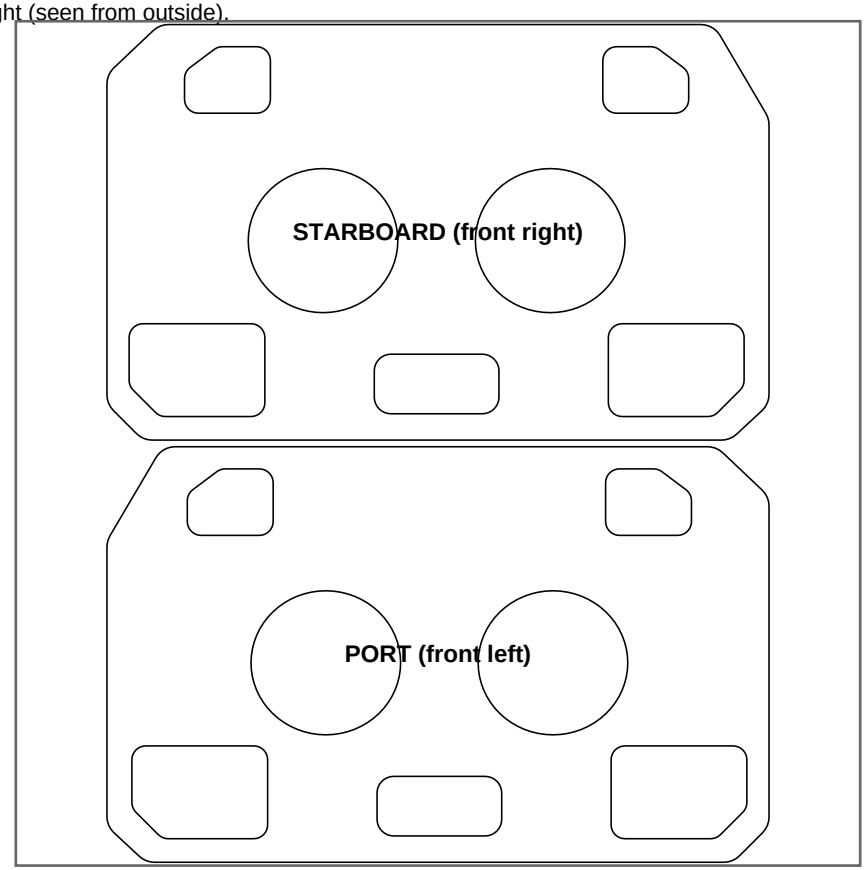
4 Centre-punch every crosshair, then drill: 3.5 mm (M3), 4.5 mm (M4), 5.5 mm (M5) as printed beside each hole. Drill straight (drill press if you have one), slow speed, back the sheet with scrap wood.

5 Drill the 10 mm blue starter hole in each window, then cut the windows with a jigsaw: fine-tooth blade, slow speed, so the HDPE does not melt and re-weld behind the blade. Cut just outside the line.

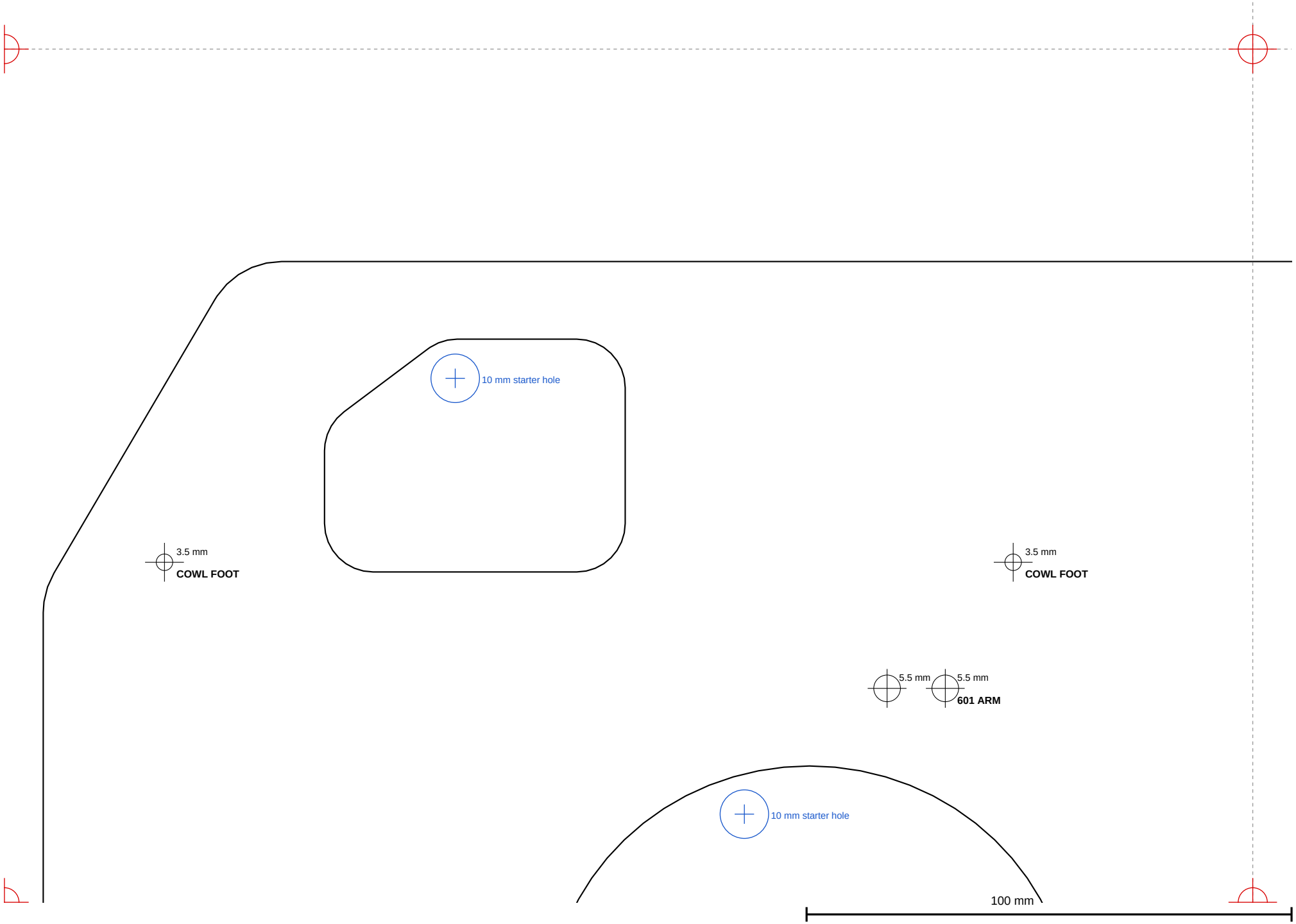
6 Cut the two plate outlines, then file / sand every edge smooth and break the sharp corners.

7 The two plates are mirror images: PORT has the front of the ROV on the left, STARBOARD on the right (seen from outside). Mark each plate's outside face before you peel the template off.

Tools: jigsaw + fine blades, drill + 3.5 / 4.5 / 5.5 / 10 mm bits, centre punch, clamps, file or sandpaper.

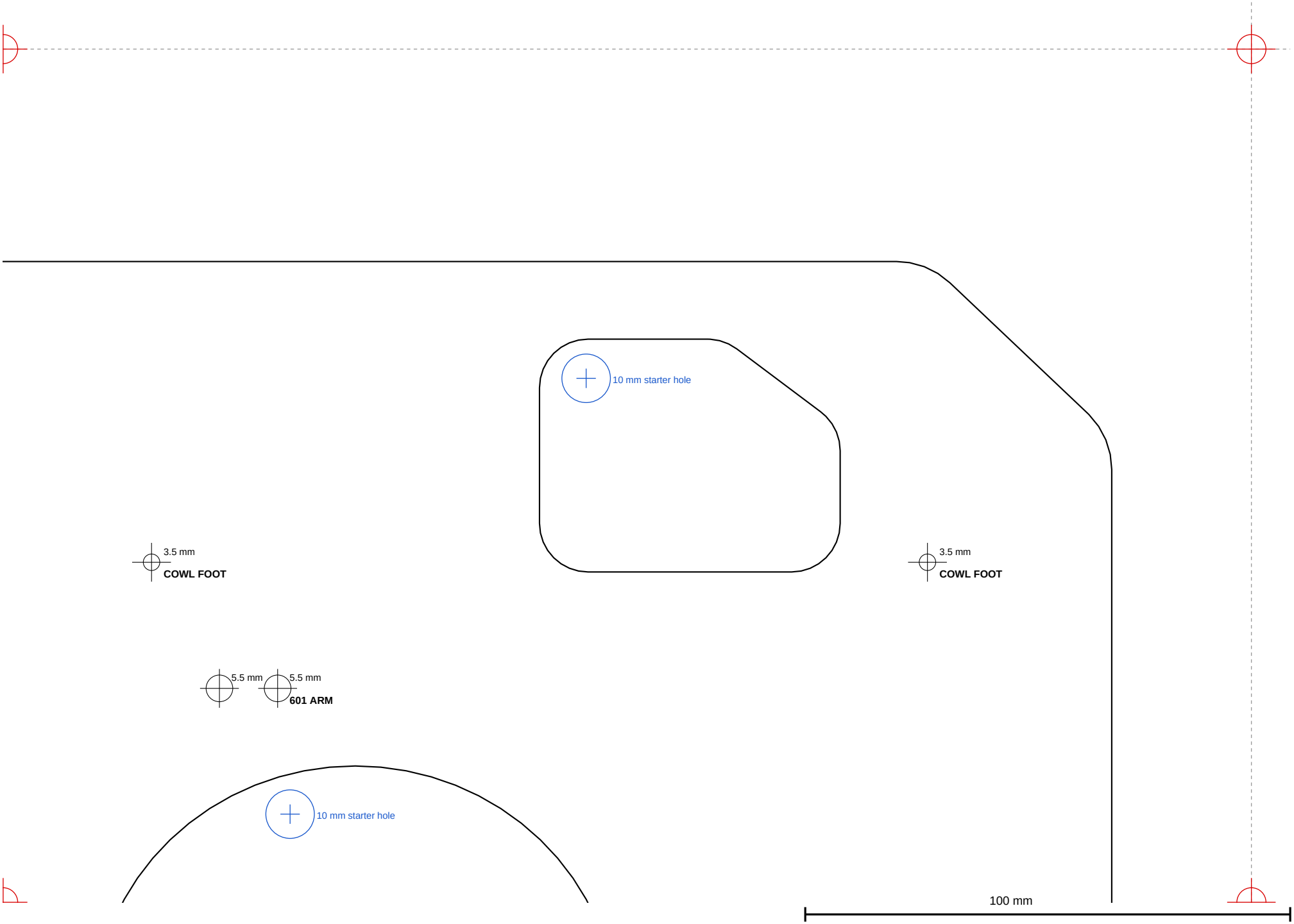


24 x 24 in sheet (609.6 mm), both plates outside face up



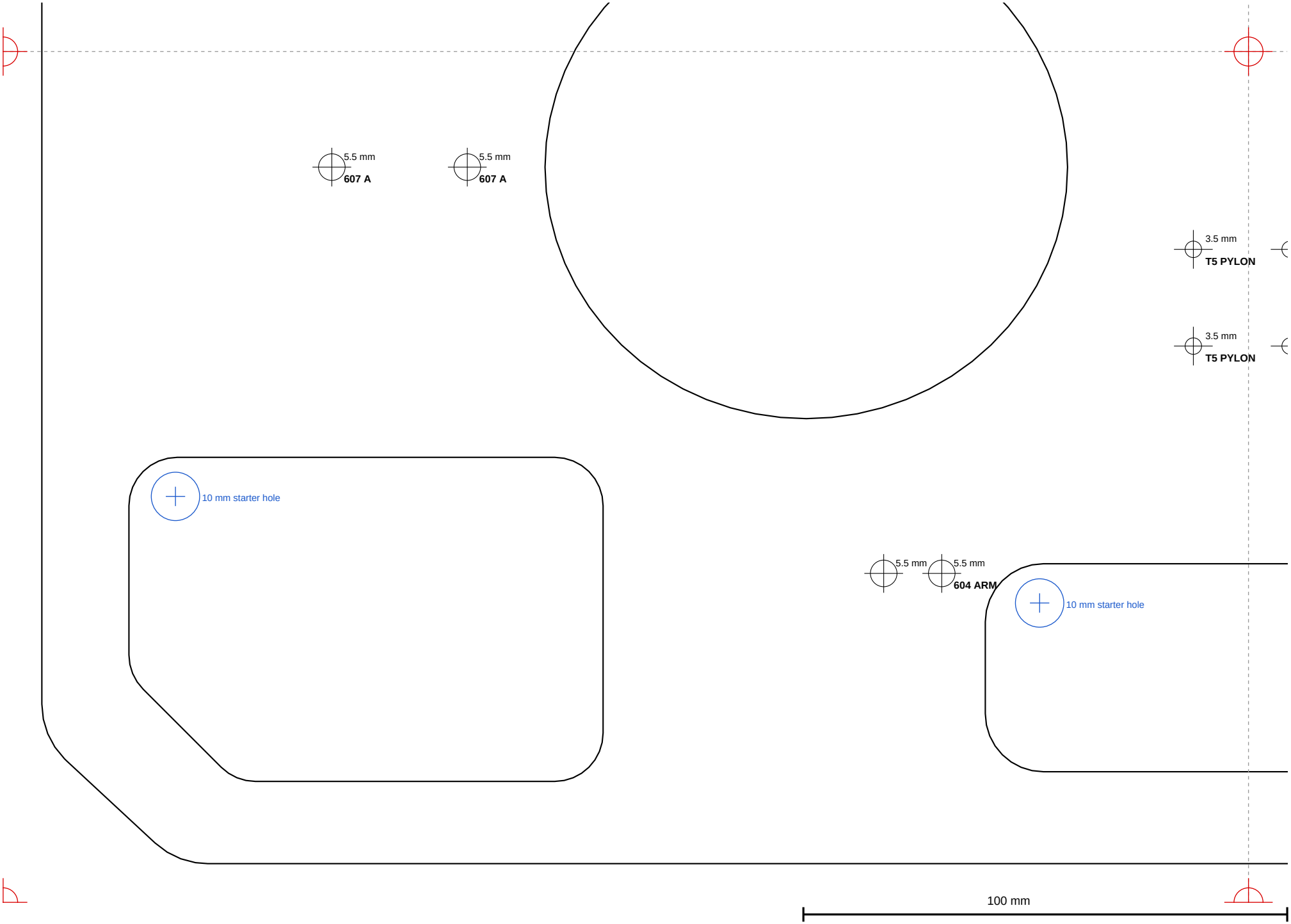
TinkerSub KB-1 side plate 901 PORT - tile A1 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



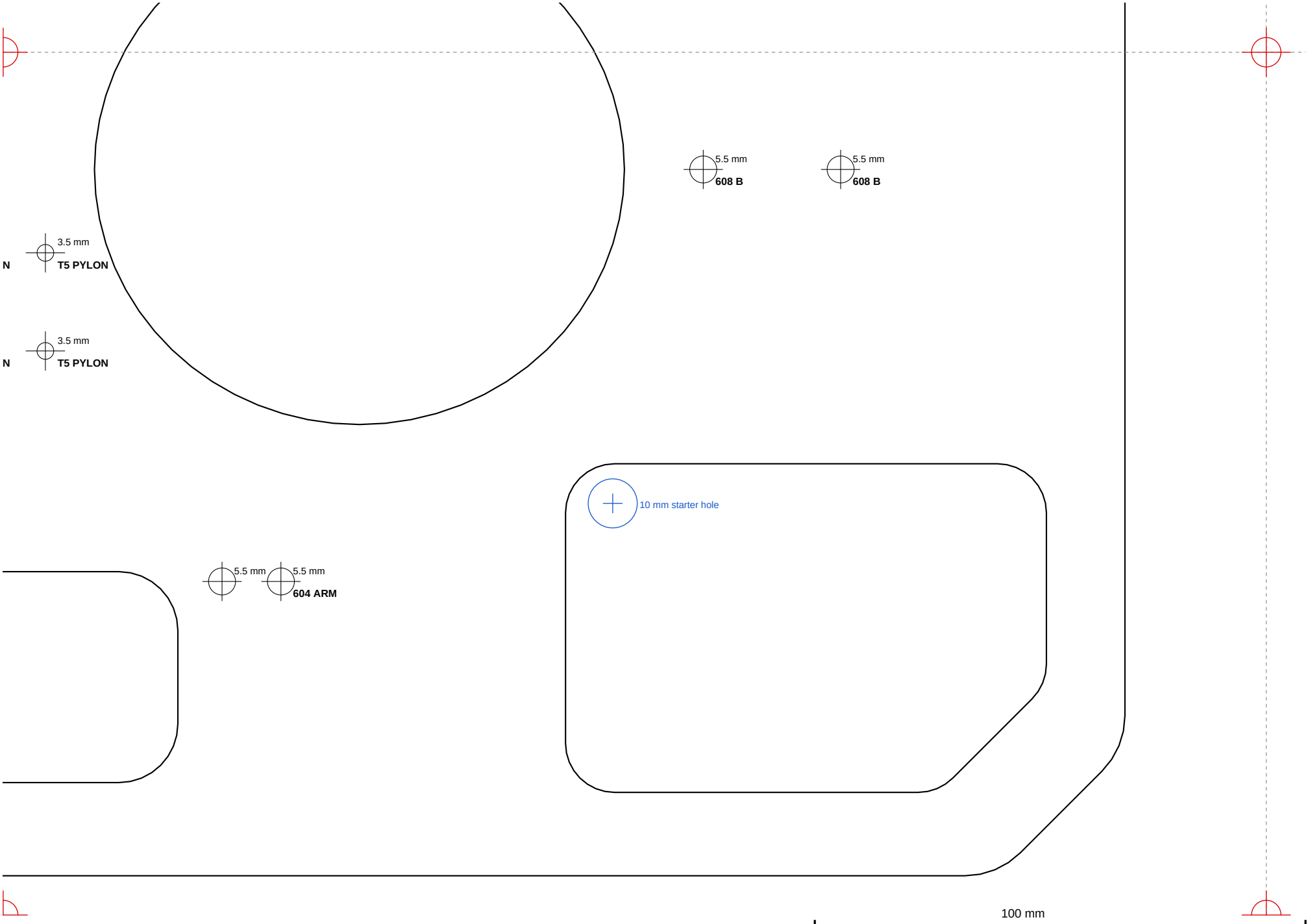
TinkerSub KB-1 side plate 901 PORT - tile A2 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



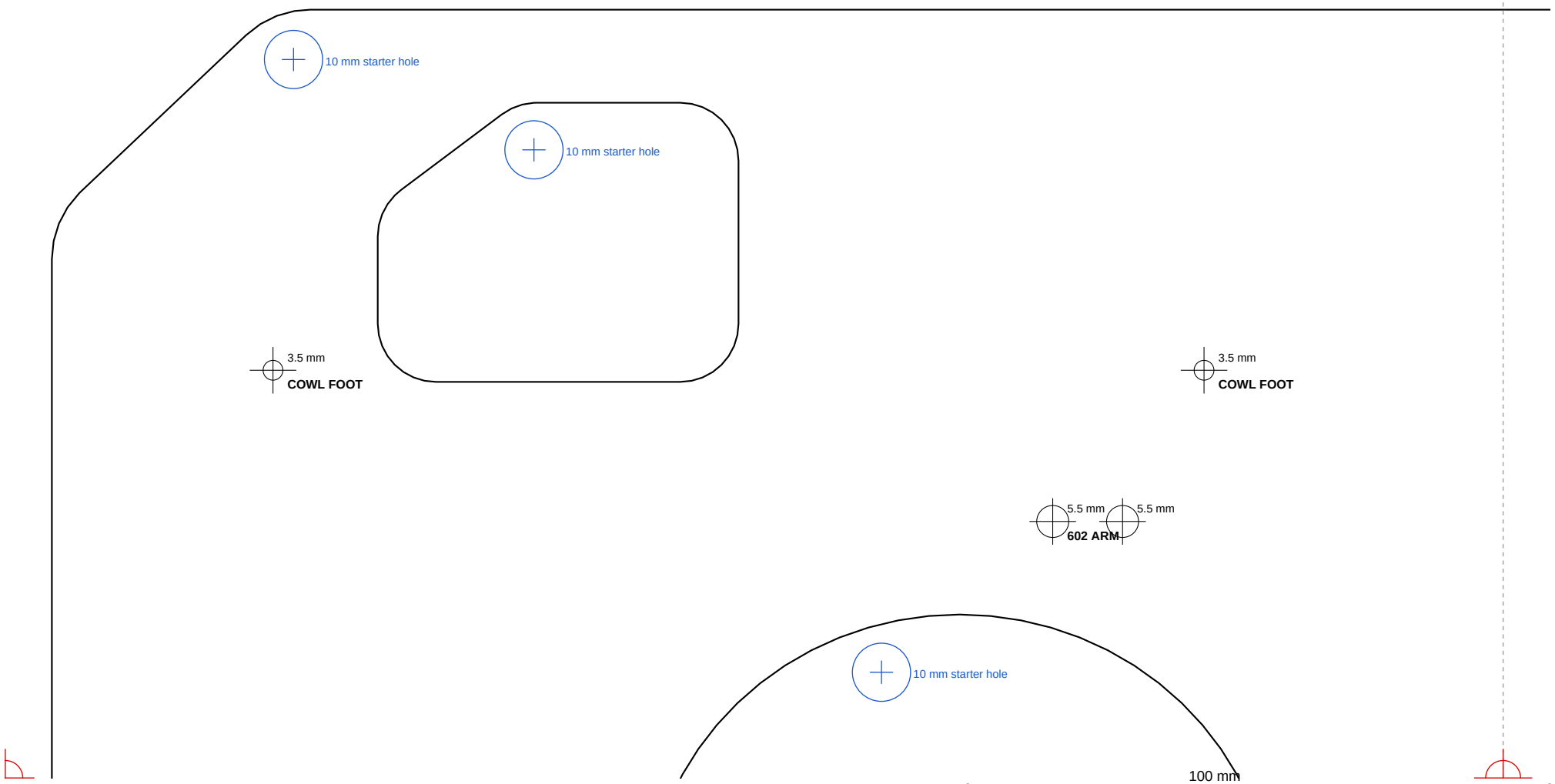
TinkerSub KB-1 side plate 901 PORT - tile B1 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



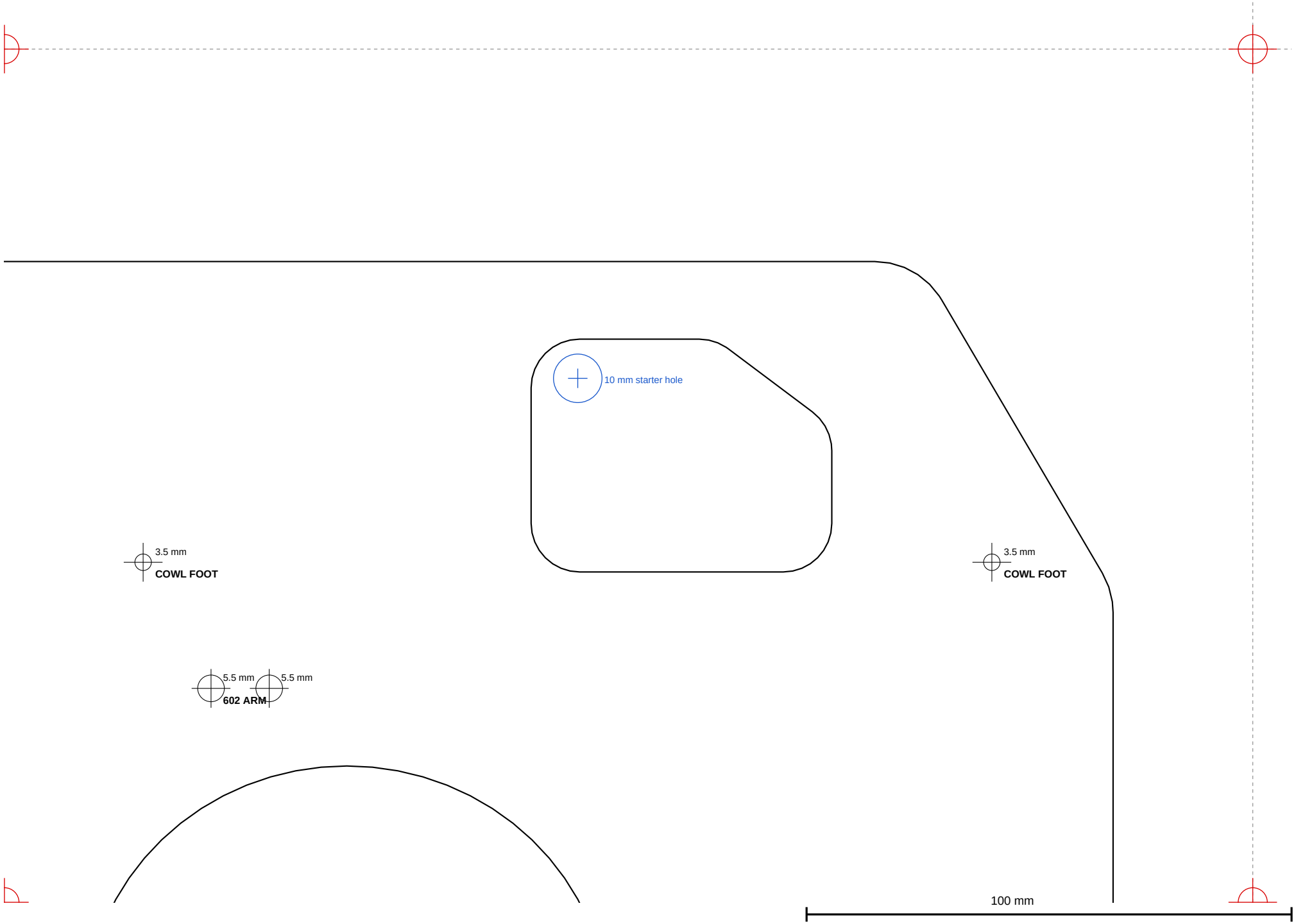
TinkerSub KB-1 side plate 901 PORT - tile B2 of 4 - seen from OUTSIDE, front of the ROV to the left

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



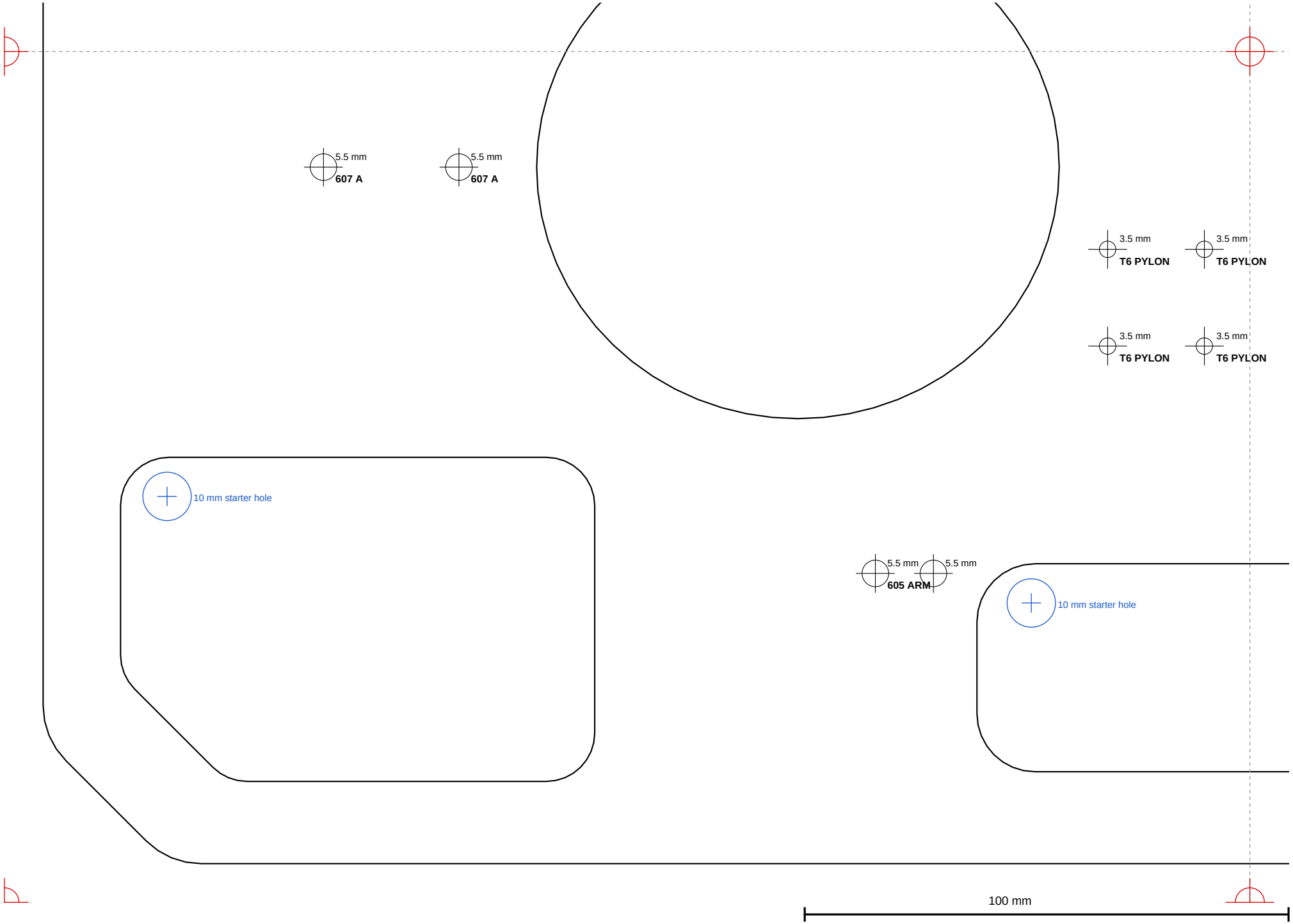
TinkerSub KB-1 side plate 902 STARBOARD - tile A1 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



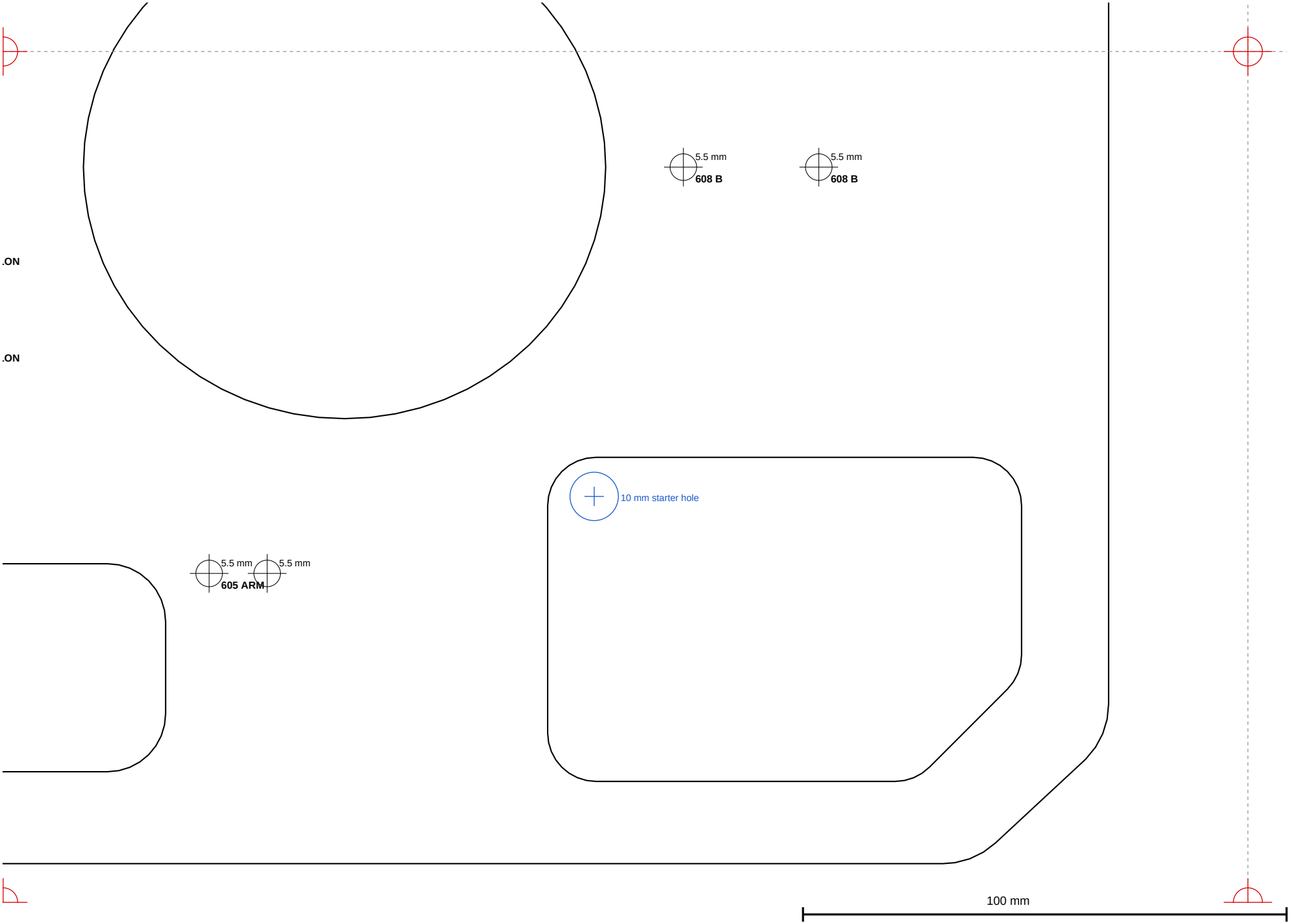
TinkerSub KB-1 side plate 902 STARBOARD - tile A2 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



TinkerSub KB-1 side plate 902 STARBOARD - tile B1 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.



TinkerSub KB-1 side plate 902 STARBOARD - tile B2 of 4 - seen from OUTSIDE, front of the ROV to the right

Print at 100 % / Actual size (no scaling). Check the 100 mm bar. Trim one tile edge, overlap the next on the dashed line, match the red marks, tape. Holes: 3.5 mm = M3, 4.5 mm = M4, 5.5 mm = M5.